

UAV Anti-Jamming Communications With Power and Mobility Control

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Abstract—Unmanned aerial vehicle (UAV)-enabled air-ground integrated communication systems are vulnerable to jamming attack, mainly due to the high probability with line-of-sight wireless channel conditions. Although a few UAV anti-jamming transmission schemes have been recently proposed, the fundamental performance limits by simultaneously considering the UAV and jammer's mobility have not yet been reported. These observations motivate us to formulate an interactive reward region (IRR) characterization problem for the scenario that the UAV and the jammer compete with each other to maximize their respective rewards via power and mobility control. Then, we propose a transmit and jamming power optimization algorithm to characterize the IRR with power control, by leveraging successive convex approximation technique, where two transmit power design schemes are given in closed-form expressions from a game-theoretic perspective. Furthermore, the IRR with joint power and mobility control is characterized based on alternating optimization, in which both the UAV and jammer can simultaneously adjust their power and trajectories. Finally, extensive simulation results reveal that joint transmit power and trajectory optimization can enlarge the IRR, and increasing the UAV mobility and flight time length improves the UAV reward.

Index Terms—UAV communications, anti-jamming, reward region, power control, trajectory optimization.

I. INTRODUCTION

A. Background and Motivation

RECENTLY, unmanned aerial vehicle (UAV) has attracted much attention owing to its flexible deployment and

controllable mobility. With higher probability of having line-of-sight (LoS) communication links [1], UAVs can boost the capacity and coverage of existing terrestrial networks, especially in areas where communication nodes are densely deployed or the existing infrastructure is damaged [2]. Compared with terrestrial base stations, one of advantages by leveraging UAVs in wireless communications is that they can proactively construct favorable channels via mobility control to provide an additional degree of freedom for system performance improvement [3].

However, as the probability of LoS transmissions increases and spatial positions are controllable, the interference from other transmitters also becomes more serious and thus the UAV systems are more vulnerable to jamming attack. This motivates the studies on UAV anti-jamming communications in the case that a jamming attacker prevents devices from communicating by disrupting the reception of communications at the legitimate receiver [4]. The anti-jamming transmission taking into account the UAV characteristics is one of the key design challenges for UAV communications. Besides traditional anti-jamming mechanisms that deal with malicious jamming via transmit power optimization [5], anti-jamming transmissions for UAV communications can be also realized by trajectory optimization [6], [7] or altitude design [8]. Moreover, to characterize the interactions between the UAV and jammer, game theoretical approaches are generally applied, where each player attempts to maximize its reward (utility).

The existing studies provide lots of insightful ideas for UAV anti-jamming communications, where players' rewards are realized at an equilibrium point. In the non-cooperative scenario, the maximum rewards that can be achieved by different anti-jamming methods are more concerned, i.e., the interactive reward region (IRR). The IRR can intuitively evaluate the performance of anti-jamming methods. However, the basic question that “*What is the IRR of UAV anti-jamming communications via resource allocation and trajectory optimization?*” is an imperative but unsolved problem.

Although characterizing the reward regions of anti-jamming transmissions is insightful for the system design, many challenges are also faced. In practice, both the UAV and jammer are intelligent agents, i.e., the UAV can learn the optimal anti-jamming strategy. On the other hand, the jammer adjusts its jamming strategy according to the UAV's behaviour. Thus, the strategy selection between the UAV and jammer is a dynamic game, where there are infinite strategies and the strategies are changing in real time. The main challenge to answer the above question lies in that how the proposed reward

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region characterization method covers all anti-jamming and jamming strategies. This requires us to construct a generalized anti-jamming transmission scenario, where neither the UAV or the jammer has asymmetric ability. These observations motivate us to present a basic model and develop fundamental results on the IRR for the scenario where the UAV and jammer compete with each other to maximize their rewards via power optimization and trajectory planning.

B. Related Work

Although UAVs can enhance existing wireless networks, a number of issues need to be addressed such as limited onboard energy and dynamic network topology. To address these challenges, lots of studies on UAV communications have been reported recently for improving wireless coverage [9], [10], instantaneous or average throughput [11], [12], and energy consumption [13], [14], etc. The existing studies have also covered other issues related to UAV communications, including mobile edge computing [15], caching [16], and intelligent reflecting surface systems [17]. Besides, the studies of UAV-to-users links have been conducted in various existing work. For example, it is shown in [18] and [19] that the performance of the UAV-assisted radio frequency/free-space optical system is significantly influenced by the shadowing effect and multi-path effect on the radio frequency link. These efforts have proved that introducing the UAV into wireless communication networks can enhance the system performance, generally via resource allocation and trajectory optimization. The performance improvements benefit from the UAV mobility, which is controlled to reduce the distance and enhance the channel quality between the UAV and users.

The anti-jamming transmission for UAV communications can be also realized by trajectory optimization. By adjusting the spatial positions of UAVs, the signal strength of jammers can be effectively reduced. In particular, an anti-jamming three-dimensional trajectory design scheme is proposed in [7] in the presence of a malicious ground jammer with fixed locations. Simulation results demonstrate the significance of trajectory design for anti-jamming transmissions. To reflect the non-cooperative characteristics of jammers, imperfect information (e.g., uncertain locations) of jammers are considered in [20] and [21]. The worst case method is introduced to address location uncertainties of terrestrial transmitters, which guarantees that the obtained solution is feasible for all cases. However, the aforementioned studies [6], [7], [20], [21] assume that the UAV and jammer have asymmetric ability, e.g., the jammer has limited mobility while the UAV has high mobility, which makes the results not promise to support for general scenarios.

Generally, the UAV and jammer as independent and intelligent players have similar communication and mobility performance. In [22], an anti-jamming UAV relay game is formulated in which the UAV decides whether or not to relay the message, and the jammer chooses its jamming power accordingly. The UAV reward is denoted by the signal-to-interference-plus-noise-ratio (SINR) and the transmit cost, while the jammer reward is related to the energy consumption of the UAV and the jamming cost. Similarly, the authors in [23]

present a UAV aided cellular framework against jamming, in which the UAV uses reinforcement learning to choose the relay policy for a mobile user. Considering the unknown or dynamic system information, reinforcement learning based methods have the ability to learn from the interactions with the environment [24], [25]. The proposed deep Q-learning-based UAV power allocation method in [26] can suppress the attack motivation of subjective smart attackers, where the SINR and energy consumption are also considered for users' rewards. From a Stackelberg game perspective, the authors in [28] propose a trajectory optimization strategy against intelligent UAV jamming based on deep Q-network. The mobility cost per unit distance is introduced in reward formulation. A Bayesian Stackelberg game is proposed in [29] to formulate the competitive relations between UAVs (users) and the jammer. Based on utility functions, the jammer and users select their optimal power control strategies respectively.

C. Contribution

The main contribution of this paper is to present a basic model and develop fundamental results on the IRR for the scenario where the UAV and jammer compete with each other to maximize their rewards via power optimization and trajectory planning. Specifically, the contributions of this paper are summarized as follows:

- We formulate the IRR characterization problem considering the power and mobility constraints of the UAV and jammer. The rewards depend on the transmission rate, as well as energy costs of transmit power and mobility. Based on the concept of rate profile, the IRR boundary can be obtained by solving a series of optimization problems. The upper bound of the IRR is derived for the sake of comparison.
- We propose a power optimization algorithm to characterize the IRR with power control, by leveraging successive convex approximation (SCA) technique. Moreover, two transmit power design schemes from a game-theoretic perspective are presented. The interactions between the UAV and jammer are formulated as a Stackelberg game, and the optimal solutions are derived in closed-form expressions.
- We develop a reward region characterization algorithm with joint power and mobility control, in which the UAV and jammer can dynamically adjust their trajectories. An alternating optimization algorithm is proposed by iteratively optimizing one of the power or trajectory with the other variable being fixed. For each subproblem, the SCA technique is leveraged to address the non-convex optimization.

The IRR is obtained based on a generalized scenario in which all the anti-jamming and jamming strategies are included. More specifically, we have not only considered the situation where the UAV and jammer are completely competitive, but also involved the situation where they can cooperate. The "cooperation" herein means that anti-jamming strategies are carefully designed to misguide the jammer, which seems a collaborative scene. In addition, the UAV

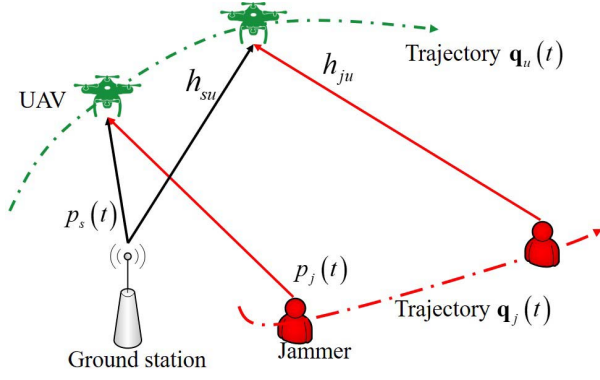


Fig. 1. Anti-jamming transmission for a UAV communication system, where the transmit power of ground station and UAV trajectory can be reconfigured.

and jammer can be smart agents with similar communication and mobility performance, or the strong one can completely suppress the other with inferior ability. The obtained IRR provides insightful guide for strategy design of UAV anti-jamming communications. The proposed IRR characterization method can be applied into confrontation scenarios, such as the case that the cellular-UAV system is attacked by illegal users. It is expected for the UAV system to mitigate the jamming from illegal users by carefully designing the UAV's behaviors and optimizing the resource allocation.

The rest of this paper is organized as follows. In Section II, we present the system model and IRR characterization problem. The IRR characterization and strategy design methods with power control are proposed in Section III. In Section IV, we characterize the IRR with the consideration of joint power control and mobility optimization. Simulation results are presented to show the IRR under different settings in Section V. Finally, we conclude this paper in Section VI.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a UAV anti-jamming communications system where a ground station communicates with an aerial UAV, e.g., controlling the UAV flight and sending or receiving surveillance data. A malicious jammer (J) with smart and programmable radio device on the ground attends to interrupt legitimate communications from the ground station to the UAV, as shown in Fig. 1. The smart jammer can jam the whole bandwidth with barrage jamming mode [28]. Moreover, the jammer can dynamically change its location and adjust its jamming power to effectively destroy the legitimate communications. Accordingly, the UAV moves freely in space with a constant altitude H , and the transmit power of ground station can be reconfigured to tackle with the smart jammer.

The UAV coordinate at any time instant t projected on the horizontal plane is denoted by $\mathbf{q}_u(t) = \{x_u(t), y_u(t)\}$. Let $h_{su}(t)$ and $h_{ju}(t)$ represent the channel coefficients from ground station to the UAV, from the jammer to the UAV, respectively. The channel coefficient $h_{su}(t)$ is characterized by the large scale pathloss $\beta_{su}(t)$ and small scale fading $\rho_{su}(t)$, i.e., $h_{su}(t) = \sqrt{\beta_{su}(t)}\rho_{su}(t)$ [31]. Let $d_{su}(t)$ denote

the distance between ground station and the UAV at time instant t . Then, the large scale channel attenuation $\beta_{su}(t)$ is $\beta_{su}(t) = \beta_0 d_{su}^{-a}(t) = \beta_0 (H^2 + \|\mathbf{q}_u(t) - \mathbf{q}_s\|^2)^{-a/2}$, where β_0 denotes the reference channel power gain at 1 m, $\mathbf{q}_s$ is the location of ground station, and $a \geq 2$ is the path loss exponent. Without loss of generality, small scale fading $\rho_{su}(t)$ is assumed to be independent and identically distributed (i.i.d.) random variables with $E[|\rho_{su}(t)|^2] = 1$. With the existence of LoS path in most scenarios, the small-scale fading is modelled by Rician fading with Rician factor K_c , i.e., $\rho_{su}(t) = \sqrt{\frac{K_c}{K_c+1}}\bar{g} + \sqrt{\frac{1}{K_c+1}}\tilde{g}$, where $\bar{g}$ denotes the deterministic LoS channel component with $|\bar{g}| = 1$, and $\tilde{g}$ denotes the random scattered component, which is a zero-mean unit-variance circularly symmetric complex Gaussian (CSCG) random variable [31]. Thus, the variance of $\rho_{su}(t)$ is $\text{Var}(\rho_{su}(t)) = \frac{1}{K_c+1}$. Similarly, the channel coefficient from the jammer to the UAV $h_{ju}(t)$ is given as $h_{ju}(t) = \sqrt{\beta_{ju}(t)}\rho_{ju}(t)$ with $\beta_{ju}(t) = \beta_0 (H^2 + \|\mathbf{q}_u(t) - \mathbf{q}_j(t)\|^2)^{-a/2}$, where $\mathbf{q}_j(t) = \{x_j(t), y_j(t)\}$ is the jammer location.

Considering the interference from the jammer, the instantaneous rate in bits/second/Hz (bps/Hz) at the UAV can be formulated as

$$r_{su}(t) = \log_2 \left(1 + \frac{p_s(t) |h_{su}(t)|^2}{p_j(t) |h_{ju}(t)|^2 + \sigma^2} \right) = \log_2 \left(1 + \frac{p_s(t) \beta_{su}(t) |\rho_{su}(t)|^2}{p_j(t) \beta_{ju}(t) |\rho_{ju}(t)|^2 + \sigma^2} \right), \quad (1)$$

where $p_s(t)$ and $p_j(t)$ are the transmit power of ground station and jamming power of the jammer at time instant t , and σ^2 denotes the noise power. The term $p_j(t) \beta_{ju}(t) |\rho_{ju}(t)|^2$ represents the interference induced by the jammer. Prior to the UAVs flight, only the channel distribution information (CDI) knowledge is known, i.e., imperfect channel state information (CSI) is considered. To address the imperfect CSI, the lower bound expression is adopted in trajectory and power design for obtaining the worst-case achievable rate. Thus, we are interested in the average rate defined as $E[r_{su}(t)]$. Then, the lower bound for $E[r_{su}(t)]$ can be obtained as [31]

$$E[r_{su}(t)] \geq \log_2 \left(1 + \frac{p_s(t) E[|h_{su}(t)|^2]}{p_s(t) \text{Var}(h_{su}(t)) + p_j(t) E[|h_{ju}(t)|^2] + \sigma^2} \right) = \log_2 \left(1 + \frac{p_s(t) \beta_{su}(t)}{p_s(t) \frac{\beta_{su}(t)}{K_c+1} + p_j(t) \beta_{ju}(t) + \sigma^2} \right) \triangleq R_{su}(t). \quad (2)$$

Let T denote the UAV total operation time length. The trajectories of UAV and jammer are continuous, which means that there are an infinite number of variables. To obtain a tractable optimization problem and characterize the IRR, the time length T of interest is discretized into N time slots, i.e., $T = N\delta_t$, where δ_t denotes the time slot length [3], [12]. Generally, δ_t should be sufficiently small such that locations of UAV and jammer are almost unchanged within

each time slot. When the operation time length is unknown, the path discretization scheme can be leveraged [20]. With time discretization, trajectories of UAV and jammer are given by a sequence of discrete points $\{\mathbf{q}_u[n], 1 \leq n \leq N\}$ and $\{\mathbf{q}_j[n], 1 \leq n \leq N\}$, respectively. Similarly, the transmit power over the time is also discretized into N points. The average transmission rate at the UAV within the n -th time slot can be formulated as

$$R_{su}[n] = \log_2 \left(1 + \frac{p_s[n] \beta_{su}[n]}{\frac{p_s[n] \beta_{su}[n]}{K_c + 1} + p_j[n] \beta_{ju}[n] + \sigma^2} \right), \quad (3)$$

with $\beta_{su}[n] = \beta_0 \left(H^2 + \|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 \right)^{-a/2}$ and $\beta_{ju}[n] = \beta_0 \left(H^2 + \|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2 \right)^{-a/2}$.

Let $\mathbf{q}_u[0]$ and $\mathbf{q}_j[0]$ denote the initial locations of UAV and jammer. The design of UAV communication system needs to consider the communication performance and UAV's characteristics. Different from the terrestrial base stations, UAVs have limited onboard energy. To this end, it is expected to achieve the maximum transmission rate via minimum energy consumption. However, there is a tradeoff between the energy consumption and transmission rate. Similar to [27], [28], and [30], the UAV reward is defined as a function of the average transmission rate, energy cost of transmit power and mobility, where parameters determine the proportions between the energy consumption and transmission rate. Then, the immediate reward of UAV at the n -th time slot is defined as

$$U_u[n] = R_{su}[n] - c_s p_s[n] - c_u \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|, \quad (4)$$

where c_s and c_u are unit transmit power and energy costs of the UAV system [28]. Then, the average reward of UAV is $U_u = \frac{1}{N} \sum_{n=1}^N U_u[n]$.

From the jammer perspective, the aim is to decrease the UAV reward (i.e., degrading the transmission rate $R_{su}[n]$ and depleting the UAV's energy) while keeping energy costs as small as possible. In this context, the immediate reward of jammer is defined as

$$U_j[n] = U^{\text{up}} - U_u[n] - c_{jp} p_j[n] - c_j \|\mathbf{q}_j[n] - \mathbf{q}_j[n-1]\|, \quad (5)$$

where U^{up} is a constant that guarantees a positive utility for the jammer, c_{jp} and c_j are unit transmit power and energy costs of the jammer. The average reward of jammer is $U_j = \frac{1}{N} \sum_{n=1}^N U_j[n]$.

B. Problem Statement

The aim of this paper is to characterize all the achievable reward pairs of UAV and jammer under given transmit power and mobility constraints, i.e., IRR. Mathematically, the IRR is given by

$$\mathcal{C} = \bigcup_{\substack{\mathbf{q}_u[n] \in \mathbf{Q}_u, \mathbf{q}_j[n] \in \mathbf{Q}_j \\ p_s[n] \in \mathbf{P}_s, p_j[n] \in \mathbf{P}_j}} \{(u_1, u_2) : 0 \leq u_1 \leq U_u, 0 \leq u_2 \leq U_j\}, \quad (6)$$

where

$$\mathbf{Q}_u = \{\mathbf{q}_u[n] : \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \leq V_u^{\text{max}} \delta_t, \forall n\}, \quad (7a)$$

$$\mathbf{Q}_j = \{\mathbf{q}_j[n] : \|\mathbf{q}_j[n] - \mathbf{q}_j[n-1]\| \leq V_j^{\text{max}} \delta_t, \forall n\}, \quad (7b)$$

$$\mathbf{P}_s = \{p_s[n] : 0 \leq p_s[n] \leq \bar{P}_s\}, \quad (7c)$$

$$\mathbf{P}_j = \{p_j[n] : 0 \leq p_j[n] \leq \bar{P}_j\}, \quad (7d)$$

with V_u^{max} and V_j^{max} as maximum flight speeds of UAV and jammer, $\bar{P}_s$ and $\bar{P}$ as transmit power constraints of UAV and jammer. To characterize the boundary of reward region, the concept of rate profile is leveraged as in [32] and [33]. Any point on the boundary can be obtained via solving the following optimization problem with a particular rate-profile vector $\mathbf{a} = \{\alpha_1, \alpha_2\}$:

$$\text{P1 : } \max_{u_0, \{\mathbf{q}_u[n], \mathbf{q}_j[n], p_s[n], p_j[n]\}} u_0 \quad (8a)$$

$$\text{s.t. } U_u \geq \alpha_1 u_0, \quad (8b)$$

$$U_j \geq \alpha_2 u_0, \quad (8c)$$

$$\mathbf{q}_u[n] \in \mathbf{Q}_u, \mathbf{q}_j[n] \in \mathbf{Q}_j, \quad (8d)$$

$$p_s[n] \in \mathbf{P}_s, p_j[n] \in \mathbf{P}_j. \quad (8e)$$

In optimization problem (8), the rate-profile vector $\mathbf{a}$ with $\{\alpha_k \geq 0, k = 1, 2\}$ and $\alpha_1 + \alpha_2 = 1$ denotes the target ratio between the achievable UAV reward and jammer reward. Denote u_0^* as the optimal solution of problem (P1) with a given rate-profile vector $\mathbf{a}$, where u_0^* also is the total reward of the UAV and jammer. Then, it follows that $u_0^* \cdot \mathbf{a}$ is the corresponding optimal rate-tuple. Therefore, solving problem (P1) under various $\mathbf{a}$ yields the complete IRR boundary, which does not need to be convex [33]. Before explicitly characterizing the IRR, we first provide the following proposition, which serves as an upper bound.

Proposition 1: Without transmit power and energy costs, i.e., $\{c_s, c_u, c_{jp}, c_j\} = \mathbf{0}$, the boundary of reward region is simply a triangle connecting three points $(0, 0)$, $(0, U^{\text{up}})$, and $(U^{\text{up}}, 0)$, denoted by $\mathcal{C}^{\text{up}}$.

Proof: Please refer to Appendix A. ■

It is noted that the reward region $\mathcal{C}^{\text{up}}$ provides an upper bound for $\mathcal{C}$, $\mathcal{C} \in \mathcal{C}^{\text{up}}$. Moreover, the UAV and jammer can be regarded as a constant-sum game where the rewards of players add up to a constant value U^{up} , for any strategy profile. When $U^{\text{up}} = 0$, it degrades into the well known zero-sum game [34]. For the purpose of illustration, Fig. 2 shows the reward region and its upper bound. The green and red circles represent the optimal solutions of UAV and jammer, which however is difficult to achieve in the adversarial scenario. The solutions obtained by existing anti-jamming methods are usually inside the region. The aim of this paper is to accurately present the IRR and provide guidance for anti-jamming or jamming algorithm design.

III. IRR CHARACTERIZATION AND STRATEGY DESIGN WITH POWER CONTROL

Power control is an efficient anti-jamming strategy in wireless networks [26], [27]. In this section, we first investigate the reward region with power control. The proposed power optimization algorithm is utilized to characterize the reward

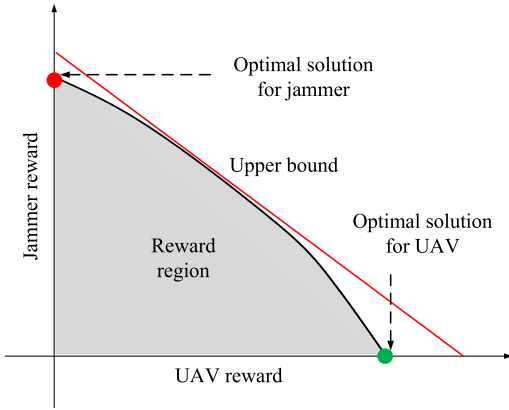


Fig. 2. Illustration of the IRR and its upper bound.

region, provided that the global information is known. However, in the confrontation scenario, it is usually difficult to obtain perfect global information for both the UAV system and the jammer. Thus, we further propose the anti-jamming transmission strategy from a game-theoretic perspective.

A. IRR Characterization With Power Control

With fixed locations $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\} = \{\mathbf{q}_u^f, \mathbf{q}_j^f\}$, $\forall n$, there are $\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| = 0$ and $\|\mathbf{q}_j[n] - \mathbf{q}_j[n-1]\| = 0$. Thus, the optimization problem (P1) can be reformulated as

$$(P2.1): \quad \max_{u_0, \{p_s[n], p_j[n]\}} u_0$$

$$\text{s.t.} \quad \sum_{n=1}^N R_{su}[n] \geq N\alpha_1 u_0 + \sum_{n=1}^N c_s p_s[n], \quad (9a)$$

$$\sum_{n=1}^N (R_{su}[n] + c_{jp} p_j[n] - c_s p_s[n]) \leq NU^{\text{up}} - N\alpha_2 u_0, \quad (9b)$$

$$p_s[n] \in \mathbf{P}_s, p_j[n] \in \mathbf{P}_j. \quad (9c)$$

The problem (P2.1) is nonconvex due to (9a) and (9b), and thus standard convex optimization techniques cannot be directly used. For simplicity, denote

$$\gamma_1[n] = \frac{p_s[n] \beta_{su}}{(K_c + 1)} + p_j[n] \beta_{ju} + \sigma^2, \quad (10a)$$

$$\gamma_2[n] = p_s[n] \beta_{su} + \frac{p_s[n] \beta_{su}}{(K_c + 1)} + p_j[n] \beta_{ju} + \sigma^2. \quad (10b)$$

Notice that β_{su} and β_{ju} are given with fixed locations, and thus we omit the index n . Then, the transmission rate $R_{su}[n]$ at the UAV can be expressed as $R_{su}[n] = \log_2 \gamma_2[n] - \log_2 \gamma_1[n]$, where $\log_2 \gamma_1[n]$ and $\log_2 \gamma_2[n]$ are jointly concave with respect to $\{p_s[n], p_j[n]\}$. Therefore, with given local points $\{p_s^k[n], p_j^k[n]\}$, we can obtain the following upper bounds for $\log_2 \gamma_1[n]$ and $\log_2 \gamma_2[n]$ by applying the first-order Taylor expansion:

$$\log_2 \gamma_1[n] \leq \log_2 \gamma_1^k[n] + \frac{\beta_{su} \log_2 e}{\gamma_1^k[n] (K_c + 1)} (p_s[n] - p_s^k[n])$$

$$+ \frac{\beta_{ju} \log_2 e}{\gamma_1^k[n]} (p_j[n] - p_j^k[n])$$

$$\triangleq f_{p1}(p_s[n], p_j[n]), \quad (11a)$$

$$\log_2 \gamma_2[n] \leq \log_2 \gamma_2^k[n] + \frac{(K_c + 2) \beta_{su} \log_2 e}{\gamma_2^k[n] (K_c + 1)} (p_s[n] - p_s^k[n])$$

$$+ \frac{\beta_{ju} \log_2 e}{\gamma_2^k[n]} (p_j[n] - p_j^k[n])$$

$$\triangleq f_{p2}(p_s[n], p_j[n]). \quad (11b)$$

It can be easily observed that $f_{p1}(p_s[n], p_j[n])$ and $f_{p2}(p_s[n], p_j[n])$ are affine functions of $\{p_s[n], p_j[n]\}$. Considering the fact that $R_{su}[n] = \log_2 \gamma_2[n] - \log_2 \gamma_1[n]$, there is

$$\log_2 \gamma_2[n] - f_{p1}(p_s[n], p_j[n]) \leq R_{su}[n]$$

$$\leq f_{p2}(p_s[n], p_j[n]) - \log_2 \gamma_1[n], \quad (12)$$

in which the throughput of UAV system is bounded by $\log_2 \gamma_2[n] - f_{p1}(p_s[n], p_j[n])$ and $f_{p2}(p_s[n], p_j[n]) - \log_2 \gamma_1[n]$.

In the context of (12), if the following constraints are satisfied,

$$\sum_{n=1}^N [\log_2 \gamma_2[n] - f_{p1}(p_s[n], p_j[n])] \geq N\alpha_1 u_0 + \sum_{n=1}^N c_s p_s[n], \quad (13a)$$

$$\sum_{n=1}^N (f_{p2}(p_s[n], p_j[n]) - \log_2 \gamma_1[n] + c_{jp} p_j[n] - c_s p_s[n]) \leq NU^{\text{up}} - N\alpha_2 u_0, \quad (13b)$$

then, the constraints in (9a) and (9b) can also be guaranteed. Therefore, the optimization problem (P2.1) can be solved by iteratively addressing the following convex problem:

$$(P2.2): \quad \max_{u_0, \{p_s[n], p_j[n]\}} u_0$$

$$\text{s.t.} \quad (9c), (13). \quad (14)$$

Specifically, initializing from a point $\{p_s[n], p_j[n]\}^{(0)}$, we can get a sequence of non-decreasing solutions with iteratively updating the local points $\{p_s[n], p_j[n]\}^{(k)} = \{p_s[n], p_j[n]\}^{(k+1)}$. It can be proved that the convergence is guaranteed for the SCA algorithm [12]. With various $\mathbf{a} = \{\alpha_1, \alpha_2\}$, the complete reward region achieved by power control can be obtained, as shown in Algorithm 1.

B. Transmit Power Design

In this subsection, we investigate the transmit power design scheme from a game-theoretic perspective. The power control strategies of UAV and jammer are analyzed under the Stackelberg game framework, where one acts as leader and another acts as follower [35]. The power control game is denoted by $\mathcal{G} = \{\langle u, j \rangle, \langle p_s, p_j \rangle, \langle U_s, U_j \rangle\}$, in which the players are UAV (u) and jammer (j), the actions $\langle p_s, p_j \rangle$ of the players follow the power constraints, and $\langle U_s, U_j \rangle$ are their rewards.

Generally, each player in a Stackelberg game is assumed to be rational and selfish with the aim of maximizing its reward. The leader chooses its strategy before the follower

Algorithm 1 IRR Characterization With Power Control

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1: For various  $\{\alpha_1, \alpha_2\}$ 
2:   Initialize  $\{p_s[n], p_j[n]\}^0$ , and  $k = 0$ 
3:   Repeat
4:     Solve (P2.2) by standard convex optimization techniques
       and obtain the optimal solution  $\{p_s[n], p_j[n]\}^*$ 
5:     Update the transmit power with the optimal solution
        $\{p_s[n], p_j[n]\}^{(k+1)} = \{p_s[n], p_j[n]\}^*$ 
6:     Update  $k = k + 1$ 
7:   Until some termination conditions are met
8: EndFor

```

decides its strategy such that the follower can observe the leader's strategy and adapt its strategy accordingly. On the other hand, the leader is aware of the fact that the follower will choose its best response to the leader's strategy. The solution of the game is known as the Stackelberg equilibrium. The Stackelberg equilibrium of the power control game $\mathcal{G}$ under the power constraints, denoted by $(p_s^{\text{SE}}, p_j^{\text{SE}})$, is given by

$$p_s^{\text{SE}} = \arg \max_{p_s} U_u(p_s, p_j), \quad (15a)$$

$$p_j^{\text{SE}} = \arg \max_{p_j} U_j(p_s, p_j). \quad (15b)$$

We first study the case (defined as SE-I) that the UAV as the leader transmits with power p_s , and then the jammer as the follower chooses its jamming power p_j . Motivated by (13), the utility functions of UAV and jammer are denoted by

$$U_u(p_s, p_j) = \log_2 \gamma_2 - f_{p1}(p_s, p_j) - c_s p_s, \quad (16a)$$

$$U_j(p_s, p_j) = U^{\text{up}} + \log_2 \gamma_1 - f_{p2}(p_s, p_j) + c_s p_s - c_{jp} p_j, \quad (16b)$$

in which $U_u(p_s, p_j)$ and $U_j(p_s, p_j)$ are the lower bounds of UAV and jammer reward, i.e., equations (4) and (5).

For the game $\mathcal{G}$ in which the UAV acts as leader, the optimal solution of the jammer is given by the following lemma.

Lemma 1: The optimal jamming power of the jammer is given by equation (17), shown at the bottom of the page.

Proof: Please refer to Appendix B. ■

With Lemma 1, the optimal solution to (15) can be obtained in closed-form. It is noted that the jamming power decreases (increases) with increasing (decreasing) transmit power of the ground station. Substituting p_j in (16a) with the optimal transmit power in (17), the UAV's utility $U_u(p_s)$ can be

expressed as

$$U_u(p_s) = \begin{cases} \log_2 \left(\frac{K_c + 2}{K_c + 1} p_s \beta_{su} + \bar{P}_j \beta_{ju} + \sigma^2 \right) \\ - \left[\frac{\beta_{su} \log_2 e}{\gamma_1^k (K_c + 1)} + c_s \right] p_s + c_2, \Pi_1, \\ \log_2 \left(\frac{K_c + 2}{K_c + 1} p_s \beta_{su} + \sigma^2 \right) \\ - \left[\frac{\beta_{su} \log_2 e}{\gamma_1^k (K_c + 1)} + c_s \right] p_s + c_1, \Pi_2, \\ \log_2 \left(p_s \beta_{su} + \frac{\gamma_2^k \beta_{ju}}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} \right) \\ - c_s p_s + c_3, \text{otherwise,} \end{cases}$$

where $c_0 = \frac{\beta_{su} \log_2 e}{\gamma_1^k (K_c + 1)} p_s^k - \log_2 \gamma_1^k$, $c_1 = c_0 + \frac{\beta_{ju} \log_2 e}{\gamma_1^k}$, $p_j^k c_2 = c_0 - \frac{\beta_{ju} \log_2 e}{\gamma_1^k} (\bar{P}_j - p_j^k)$, and $c_3 = c_0 - \frac{\beta_{ju} \log_2 e}{\gamma_1^k} \left(\frac{\gamma_2^k}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} - \frac{\sigma^2}{\beta_{ju}} - p_j^k \right)$. It can be observed that there is no convexity or monotonicity guarantee for $U_u(p_s)$. The following Lemma 2 provides the optimal solution of problem (15a) in closed-form.

Lemma 2: The optimal transmit power of the UAV is given by

$$p_s^* = \arg \max_{p_s \in \{0, p_s^1, p_s^2, \bar{p}_s^1, \bar{p}_s^2, \bar{p}_s^3, \bar{P}_s\} \cap \bar{\mathbf{P}}_s} U_u(p_s), \quad (18)$$

where

$$\bar{p}_s^1 = \frac{(K_c + 1) \gamma_1^k}{\beta_{su} + (K_c + 1) \gamma_1^k c_s \ln 2} - \frac{(K_c + 1) (\bar{P}_j \beta_{ju} + \sigma^2)}{(K_c + 2) \beta_{su}}, \quad (19a)$$

$$\bar{p}_s^2 = \frac{(K_c + 1) \gamma_1^k}{\beta_{su} + (K_c + 1) \gamma_1^k c_s \ln 2} - \frac{(K_c + 1) \sigma^2}{(K_c + 2) \beta_{su}}, \quad (19b)$$

$$\bar{p}_s^3 = \frac{\log_2 e}{c_s} - \frac{\gamma_2^k \beta_{ju}}{(\beta_{ju} + c_{jp} \gamma_2^k \ln 2) \beta_{su}}, \quad (19c)$$

and $\bar{\mathbf{P}}_s = \{0 \leq \bar{p}_s^1 \leq p_s^1 \leq \bar{p}_s^3 \leq p_s^2 \leq \bar{p}_s^2 \leq \bar{P}_s\}$.

Proof: Please refer to Appendix C. ■

It is noted that $p_s^* \in \{0, p_s^1, p_s^2, \bar{p}_s^1, \bar{p}_s^2, \bar{p}_s^3, \bar{P}_s\}$, and we just calculate and compare their values at several points to obtain the optimal solution, which sharply reduces the computational complexity. Moreover, in special case where $p_s^1 \leq 0$ and $p_s^2 \geq \bar{P}_s$, there are $U_u(0) = \log_2 \left(\frac{\gamma_2^k \beta_{ju}}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} \right) + c_3$ and $U_u(\bar{P}_s) = \log_2 \left(\bar{P}_s \beta_{su} + \frac{\gamma_2^k \beta_{ju}}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} \right) - c_s \bar{P}_s + c_3$.

In hierarchical decision-making game, e.g., Stackelberg game, the leading player generally is in a strong position to enforce its strategy upon the other players. To further analyze

$$p_j = \begin{cases} \bar{P}_j, & \Pi_1 : p_s \leq p_s^1 = \left(\frac{\beta_{ju}}{\beta_{ju} / \gamma_2^k + c_{jp} \ln 2} - \beta_{ju} \bar{P}_j - \sigma^2 \right) \frac{K_c + 1}{\beta_{su}}, \\ 0, & \Pi_2 : p_s \geq p_s^2 = \left(\frac{\beta_{ju}}{\beta_{ju} / \gamma_2^k + c_{jp} \ln 2} - \sigma^2 \right) \frac{K_c + 1}{\beta_{su}}, \\ \frac{\gamma_2^k}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} - \frac{(K_c + 1) \sigma^2 + p_s \beta_{su}}{(K_c + 1) \beta_{ju}}, & \text{otherwise} \end{cases} \quad (17)$$

the behaviors of UAV and jammer, we investigate the case (defined as SE-II) that the jammer acts as leader and UAV acts as follower. The optimal solution is given by Lemma 3.

Lemma 3: For the power control game with jammer as leader and UAV as follower, the optimal solutions are given by

$$p_j^* = \arg \max_{p_j \in \{0, p_j^1, p_j^2, \tilde{p}_j^1, \tilde{p}_j^2, \tilde{p}_j^3, \bar{P}_j\} \cap \bar{\mathbf{P}}_j} U_j(p_j), \quad (20)$$

and

$$p_s^* = \begin{cases} \bar{P}_s, \Pi_3 : p_j \leq p_j^1 = \frac{\gamma_1^k (K_c + 2) \beta_{su}}{\beta_{ju} \beta_{su} + c_s \gamma_1^k \beta_{ju} (K_c + 1) \ln 2} \\ \quad - \frac{\bar{P}_s (K_c + 2) \beta_{su}}{(K_c + 1) \beta_{ju}} - \frac{\sigma^2}{\beta_{ju}}, \\ 0, \Pi_4 : p_j \geq p_j^2 = \frac{\gamma_1^k (K_c + 2) \beta_{su}}{\beta_{ju} \beta_{su} + c_s \gamma_1^k \beta_{ju} (K_c + 1) \ln 2} \\ \quad - \frac{\sigma^2}{\beta_{ju}}, \\ \frac{\gamma_1^k (K_c + 1)}{\beta_{su} + c_s \gamma_1^k (K_c + 1) \ln 2} \\ \quad - \frac{(K_c + 1) (p_j \beta_{ju} + \sigma^2)}{(K_c + 2) \beta_{su}}, \text{ otherwise,} \end{cases}$$

where

$$\tilde{p}_j^1 = \frac{\gamma_2^k}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} - \frac{\bar{P}_s \beta_{su}}{\beta_{ju} (K_c + 1)} - \frac{\sigma^2}{\beta_{ju}}, \quad (21a)$$

$$\tilde{p}_j^2 = \frac{\gamma_2^k}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} - \frac{\sigma^2}{\beta_{ju}}, \quad (21b)$$

$$\tilde{p}_j^3 = \frac{(K_c + 2) \beta_{su} \log_2 e}{c_s (K_c + 1) \beta_{ju} + c_{jp} (K_c + 2) \beta_{su}} - \frac{(K_c + 2) b_3}{(K_c + 1) \beta_{ju}}. \quad (21c)$$

and

$$U_j(p_j) = \begin{cases} \log_2 \left(\frac{\bar{P}_s \beta_{su}}{(K_c + 1)} + p_j \beta_{ju} + \sigma^2 \right) \\ \quad - \left[\frac{\beta_{ju}}{\gamma_2^k \ln 2} + c_{jp} \right] p_j + b_2, \Pi_3, \\ \log_2 (p_j \beta_{ju} + \sigma^2) - \left[\frac{\beta_{ju}}{\gamma_2^k \ln 2} + c_{jp} \right] p_j + b_1, \Pi_4, \\ \log_2 \left(\frac{K_c + 1}{K_c + 2} p_j \beta_{ju} + b_3 \right) \\ \quad - \left[\frac{c_s (K_c + 1) \beta_{ju}}{(K_c + 2) \beta_{su}} + c_{jp} \right] p_j + b_4, \text{ otherwise,} \end{cases}$$

with $b_1 = U^{\text{up}} + \frac{\beta_{ju} p_j^k}{\gamma_2^k \ln 2} + \frac{(K_c + 2) \beta_{su}}{\gamma_2^k (K_c + 1) \ln 2} p_s^k - \log_2 \gamma_2^k$, $b_2 = b_1 - \frac{(K_c + 2) \beta_{su} \bar{P}_s}{\gamma_2^k (K_c + 1) \ln 2} + c_s \bar{P}_s$, $b_3 = \frac{\gamma_1^k \beta_{su}}{\beta_{su} + c_s \gamma_1^k (K_c + 1) \ln 2} + \frac{K_c + 1}{K_c + 2} \sigma^2$, and $b_4 = \frac{\gamma_1^k \gamma_2^k (K_c + 1) c_s - (K_c + 2) \gamma_1^k \beta_{su} \log_2 e}{\beta_{su} \gamma_2^k + c_s \gamma_1^k \gamma_2^k (K_c + 1) \ln 2} - \frac{c_s (K_c + 1) \sigma^2}{(K_c + 2) \beta_{su}} + \frac{\sigma^2}{\gamma_2^k \ln 2} + b_1$.

Proof: Please refer to Appendix D. ■

Similarly with Lemma 1, the optimal solution to SE-II can be obtained in closed-form.

IV. IRR CHARACTERIZATION WITH POWER AND MOBILITY CONTROL

Based on the previous results, this section aims to reveal the IRR achieved by joint power and mobility control. However,

the constraints in (8b) and (8c) are nonconvex. To this end, the instantaneous rate $R_{su}[n]$ at the UAV is rewritten as

$$R_{su}[n] = \hat{R}_{su}[n] - \hat{R}_{su}[n], \quad (22)$$

with

$$\hat{R}_{su}[n] = \log_2 \left(\frac{p_s[n] \beta_0 + \frac{p_s[n] \beta_0}{K_c + 1}}{\sigma^2 (H^2 + \|\mathbf{q}_s - \mathbf{q}_u[n]\|^2)^{\frac{\alpha}{2}}} + \frac{p_j[n] \beta_0}{\sigma^2 (H^2 + \|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2)^{\frac{\alpha}{2}}} + 1 \right), \quad (23)$$

and

$$\hat{R}_{su}[n] = \log_2 \left(\frac{\frac{p_s[n] \beta_0}{K_c + 1}}{\sigma^2 (H^2 + \|\mathbf{q}_s - \mathbf{q}_u[n]\|^2)^{\frac{\alpha}{2}}} + \frac{p_j[n] \beta_0}{\sigma^2 (H^2 + \|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2)^{\frac{\alpha}{2}}} + 1 \right). \quad (24)$$

Then, the constraints in (8b) and (8c) are given by

$$\begin{aligned} & \sum_{n=1}^N (\hat{R}_{su}[n] - \hat{R}_{su}[n]) \\ & \geq f_1[n] = N \alpha_1 u_0 \\ & \quad + \sum_{n=1}^N (c_s p_s[n] + c_u \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|), \end{aligned} \quad (25a)$$

$$\begin{aligned} & \sum_{n=1}^N (\hat{R}_{su}[n] - \hat{R}_{su}[n] - c_u \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|) \\ & \leq f_2[n] = -N \alpha_2 u_0 \\ & \quad + \sum_{n=1}^N (U^{\text{up}} + c_s p_s[n] - c_{jp} p_j[n] - c_j \|\mathbf{q}_j[n] \\ & \quad - \mathbf{q}_j[n-1]\|). \end{aligned} \quad (25b)$$

Although $\hat{R}_{su}[n]$ and $\hat{R}_{su}[n]$ are nonconvex jointly with respect to $\{\mathbf{q}_u[n], \mathbf{q}_j[n], p_s[n], p_j[n]\}$, it can be verified that both $\hat{R}_{su}[n]$ and $\hat{R}_{su}[n]$ are convex with respect to $\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2$ and $\|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2$, if transmit power $\{p_s[n], p_j[n]\}$ is provided. On the other hand, given trajectory $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}$, $\hat{R}_{su}[n]$ and $\hat{R}_{su}[n]$ are concave with respect to $\{p_s[n], p_j[n]\}$. Inspired by these observations, we propose an efficient IRR characterization method by leveraging alternating optimization and SCA techniques [36].

With given UAV and jammer locations $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}$, the transmit power optimization problem can be simplified as

$$\begin{aligned} \text{(P3.1):} \quad & \max_{u_0, \{p_s[n], p_j[n]\}} u_0 \\ & \text{s.t. (9c), (25),} \end{aligned} \quad (26)$$

which can be solved via SCA technique, as shown in Section III-A. In particular, we can obtain a series of

non-decreasing solutions via solving the following convex optimization problem:

$$\begin{aligned}
 \text{(P3.2)} : \quad & \max_{u_0, \{p_s[n], p_j[n]\}} u_0 \\
 \text{s.t.} \quad & \sum_{n=1}^N (\log_2 l_1[n] - g_{p2}[n]) \geq f_1[n], \quad (27a) \\
 & \sum_{n=1}^N (g_{p1}[n] - \log_2 l_2[n] - c_u \|\mathbf{q}_u[n] \\
 & \quad - \mathbf{q}_u[n-1]\|) \leq f_2[n], \quad (27b) \\
 & p_s[n] \in \mathbf{P}_s, p_j[n] \in \mathbf{P}_j, \quad (27c) \\
 & \quad (27d)
 \end{aligned}$$

where

$$l_1[n] = \frac{(K_c + 2) p_s[n] \beta_{su}[n]}{(K_c + 1) \sigma^2} + \frac{p_j[n] \beta_{ju}[n]}{\sigma^2} + 1, \quad (28a)$$

$$l_2[n] = \frac{p_s[n] \beta_{su}[n]}{(K_c + 1) \sigma^2} + \frac{p_j[n] \beta_{ju}[n]}{\sigma^2} + 1, \quad (28b)$$

$$\begin{aligned}
 g_{p1}[n] = & \log_2 l_1^k[n] + \frac{(K_c + 2) \beta_{su}[n]}{(K_c + 1) \sigma^2 l_1^k[n] \ln 2} (p_s[n] - p_s^k[n]) \\
 & + \frac{\beta_{ju}[n]}{\sigma^2 l_1^k[n] \ln 2} (p_j[n] - p_j^k[n]), \quad (28c)
 \end{aligned}$$

$$\begin{aligned}
 g_{p2}[n] = & \log_2 l_2^k[n] + \frac{\beta_{su}[n] \log_2 e}{(K_c + 1) \sigma^2 l_2^k[n]} (p_s[n] - p_s^k[n]) \\
 & + \frac{\beta_{ju}[n]}{\sigma^2 l_2^k[n] \ln 2} (p_j[n] - p_j^k[n]). \quad (28d)
 \end{aligned}$$

Since $\hat{R}_{su}[n]$ and $\hat{R}_{su}[n]$ are convex with respect to $\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2$ and $\|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2$ with known $\{p_s[n], p_j[n]\}$, we obtain the following lower bound for $\hat{R}_{su}[n]$ at the k -th iteration by applying the first-order Taylor expansion:

$$\begin{aligned}
 \hat{R}_{su}[n] & \geq \log_2 (J_{su}^k[n]) - I_{su}^k[n] \left(\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \right) \\
 & \quad - I_{ju}^k[n] \left(\|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \right) \\
 & \triangleq \hat{R}_{su}^l[n], \quad (29)
 \end{aligned}$$

where $\mathbf{q}_u^k[n]$ and $\mathbf{q}_j^k[n]$ are the k -th solutions and

$$\begin{aligned}
 J_{su}^k[n] = & \frac{p_s[n] \beta_0 + p_s[n] \beta_0 / (K_c + 1)}{\sigma^2 \left(H^2 + \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \right)^{a/2}} \\
 & + \frac{p_j[n] \beta_0}{\sigma^2 \left(H^2 + \|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \right)^{a/2}} + 1, \quad (30a)
 \end{aligned}$$

$$\begin{aligned}
 I_{su}^k[n] = & \frac{(p_s[n] \beta_0 + p_s[n] \beta_0 / (K_c + 1)) (a/2) \log_2 e}{J_{su}^k[n] \sigma^2 \left(H^2 + \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \right)^{a/2+1}}, \quad (30b)
 \end{aligned}$$

$$\begin{aligned}
 I_{ju}^k[n] = & \frac{p_j[n] \beta_0 (a/2) \log_2 e}{J_{su}^k[n] \sigma^2 \left(H^2 + \|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \right)^{a/2+1}}. \quad (30c)
 \end{aligned}$$

Similarly, the lower bound of $\hat{R}_{su}[n]$ is

$$\begin{aligned}
 \hat{R}_{su}[n] & \geq \log_2 \left(\hat{J}_{su}^k[n] \right) \\
 & \quad - \hat{I}_{su}^k[n] \left(\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \right) \\
 & \quad - \hat{I}_{ju}^k[n] \left(\|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \right) \\
 & \triangleq \hat{R}_{su}^l[n], \quad (31)
 \end{aligned}$$

where

$$\begin{aligned}
 \hat{J}_{su}^k[n] = & \frac{p_s[n] \beta_0 / (K_c + 1)}{\sigma^2 \left(H^2 + \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \right)^{a/2}} \\
 & + \frac{p_j[n] \beta_0}{\sigma^2 \left(H^2 + \|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \right)^{a/2}} + 1, \quad (32a)
 \end{aligned}$$

$$\begin{aligned}
 \hat{I}_{su}^k[n] = & \frac{p_s[n] \beta_0 / (K_c + 1) (a/2) \log_2 e}{\hat{J}_{su}^k[n] \sigma^2 \left(H^2 + \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \right)^{a/2+1}}, \quad (32b)
 \end{aligned}$$

$$\begin{aligned}
 \hat{I}_{ju}^k[n] = & \frac{p_j[n] \beta_0 (a/2) \log_2 e}{\hat{J}_{su}^k[n] \sigma^2 \left(H^2 + \|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \right)^{a/2+1}}. \quad (32c)
 \end{aligned}$$

Therefore, the constraints in (8b) and (8c) can be further replaced with

$$\begin{aligned}
 & \sum_{n=1}^N \hat{R}_{su}[n] \\
 & \leq \sum_{n=1}^N \hat{R}_{su}^l[n] - f_1[n], \quad (33a)
 \end{aligned}$$

$$\begin{aligned}
 & \sum_{n=1}^N \left(\hat{R}_{su}[n] - c_u \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \right) \\
 & \leq \sum_{n=1}^N \hat{R}_{su}^l[n] + f_2[n]. \quad (33b)
 \end{aligned}$$

The right-hand-sides (RHSs) of (33) are jointly concave with respect to $\mathbf{q}_u$ and $\mathbf{q}_j$. However, there is no convexity guarantee for the left-hand-sides (LHSs) of (33).

By introducing variables $\{g_{su}[n], g_{ju}[n]\}$, we have the following optimization problem:

(P3.3) :

$$\begin{aligned}
 & \max_{u_0, \{\mathbf{q}_u[n], \mathbf{q}_j[n]\}, \{g_{su}[n], g_{ju}[n]\}} u_0 \\
 \text{s.t.} \quad & \sum_{n=1}^N \hat{R}_{su}^a[n] \leq \sum_{n=1}^N \hat{R}_{su}^l[n] - f_1[n], \quad (34a)
 \end{aligned}$$

$$\begin{aligned}
 & \sum_{n=1}^N \left(\hat{R}_{su}^a[n] - c_u \|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \right) \\
 & \leq \sum_{n=1}^N \hat{R}_{su}^l[n] + f_2[n], \quad (34b)
 \end{aligned}$$

$$\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 \geq g_{su}[n], \quad \forall n, \quad (34c)$$

$$\|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2 \geq g_{ju}[n], \quad \forall n, \quad (34d)$$

$$\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \leq V_u^{\max} \delta_t, \quad \forall n, \quad (34e)$$

$$\|\mathbf{q}_j[n] - \mathbf{q}_j[n-1]\| \leq V_j^{\max} \delta_t, \quad \forall n, \quad (34f)$$

where

$$\hat{R}_{su}^a[n] = \log_2 \left(\frac{p_s[n] \beta_0 (K_c + 2) / \sigma^2}{(K_c + 1) (H^2 + g_{su}[n])^{\frac{\alpha}{2}}} + \frac{p_j[n] \beta_0 / \sigma^2}{(H^2 + g_{ju}[n])^{\frac{\alpha}{2}}} + 1 \right), \quad (35a)$$

$$\hat{R}_{su}^a[n] = \log_2 \left(\frac{p_s[n] \beta_0 / (K_c + 1)}{\sigma^2 (H^2 + g_{su}[n])^{\alpha/2}} + \frac{p_j[n] \beta_0}{\sigma^2 (H^2 + g_{ju}[n])^{\alpha/2}} + 1 \right). \quad (35b)$$

It can be shown that without loss of optimality to problem (P3.3), the constraints in (34c) and (34d) can be satisfied with equality. Otherwise, the objective value can still be improved by increasing $g_{su}[n]$ and/or $g_{ju}[n]$ while fixing other variables.

For the LHS $\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|$ of (34b), the Cauchy-Schwarz inequality $\|\mathbf{a}\| \|\mathbf{b}\| \geq |\langle \mathbf{a}, \mathbf{b} \rangle|$ is applied. For ease of presentation, denote $\mathbf{z}_u[n] = \mathbf{q}_u[n] - \mathbf{q}_u[n-1]$. Then, there is

$$\|\mathbf{z}_u[n]\| \|\mathbf{z}_u^k[n]\| \geq (\mathbf{z}_u[n])^\dagger \mathbf{z}_u^k[n], \quad (36)$$

with $\mathbf{z}_u^k[n] = \mathbf{q}_u^k[n] - \mathbf{q}_u^k[n-1]$ as the value of the k -th iteration.

Therefore, $\|\mathbf{z}_u[n]\|$ is lower bounded by

$$\|\mathbf{z}_u[n]\| \geq \frac{(\mathbf{z}_u[n])^\dagger \mathbf{z}_u^k[n]}{\|\mathbf{z}_u^k[n]\|}. \quad (37)$$

Considering the fact that $\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2$ and $\|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2$ are convex, by applying the first-order Taylor expansion at the given point $\{\mathbf{q}_j^k[n], \mathbf{q}_u^k[n]\}$, there are

$$\begin{aligned} \|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 &\geq \|\mathbf{q}_s - \mathbf{q}_u^k[n]\|^2 \\ &\quad + 2(\mathbf{q}_u^k[n] - \mathbf{q}_s)^\dagger (\mathbf{q}_u[n] - \mathbf{q}_u^k[n]) \geq g_{su}[n], \end{aligned} \quad (38a)$$

$$\begin{aligned} \|\mathbf{q}_j[n] - \mathbf{q}_u[n]\|^2 &\geq -\|\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n]\|^2 \\ &\quad + 2(\mathbf{q}_j^k[n] - \mathbf{q}_u^k[n])^\dagger (\mathbf{q}_j[n] - \mathbf{q}_u[n]) \geq g_{ju}[n]. \end{aligned} \quad (38b)$$

Thus, (P3.3) can be solved by iteratively addressing the following optimization:

$$\begin{aligned} \text{(P3.4):} \quad & \max_{u_0, \{\mathbf{q}_u[n], \mathbf{q}_j[n]\}, \{g_{su}[n], g_{ju}[n]\}} u_0 \\ & \text{s.t.} \sum_{n=1}^N \left(\hat{R}_{su}^a[n] - c_u \frac{(\mathbf{z}_u[n])^\dagger \mathbf{z}_u^k[n]}{\|\mathbf{z}_u^k[n]\|} \right) \end{aligned}$$

Algorithm 2 IRR Characterization With Joint Power and Mobility Control

```

1: For various  $\{\alpha_1, \alpha_2\}$ 
2:   Initialize  $\{\mathbf{q}_u[n], \mathbf{q}_j[n], p_s[n], p_j[n]\}^0$ , and  $k = 0$ 
3:   Repeat
4:     Set  $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\} = \{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^{(k)}$ ,  $l = 0$ ,
        $\{p_u[n], p_j[n]\}^{(l)} = \{p_u[n], p_j[n]\}^{(k)}$ 
5:     Repeat
6:       Solve (P3.2) and obtain the optimal solution
        $\{p_s[n], p_j[n]\}^*$ 
7:       Update the transmit power  $\{p_u[n], p_j[n]\}^{(l+1)} =$ 
        $\{p_u[n], p_j[n]\}^*$ 
8:       Update  $l = l + 1$ 
9:     Until some termination conditions are met
10:    Update  $\{p_u[n], p_j[n]\}^{(k)} = \{p_u[n], p_j[n]\}^{(l)}$ 
11:    Set  $\{p_u[n], p_j[n]\} = \{p_u[n], p_j[n]\}^{(k)}$ ,  $l = 0$ ,
        $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^{(l)} = \{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^{(k)}$ 
12:    Repeat
13:      Solve (P3.4) and obtain the optimal solution
        $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^*$ 
14:      Update the trajectory  $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^{(l+1)} =$ 
        $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^*$ 
15:      Update  $l = l + 1$ 
16:    Until some termination conditions are met
17:    Update  $\{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^{(k)} = \{\mathbf{q}_u[n], \mathbf{q}_j[n]\}^{(l)}$ 
18:    Until some termination conditions are met
19: EndFor

```

$$\leq \sum_{n=1}^N \hat{R}_{su}^l[n] + f_2[n], \quad (39a)$$

$$(34a), (34e), (34f), (38). \quad (39b)$$

The algorithm to characterize the IRR with joint power and mobility control is summarized in Algorithm 2. Problem (P3.4) can be efficiently solved by standard convex optimization techniques or existing solvers such as CVX [37]. The complexity is roughly calculated as $O(n^3)$ by leveraging the interior-point method, where n is the number of optimization variables. Thus, the complexity of Algorithm 1 can be estimated as about $O((6N+1)^3)$ at each iteration. From Algorithm 2, it can be observed that there are many iterations, which inevitably increases the complexity. In practice, the number of intermediate iterations (i.e., lines 5-9 and 12-16 of Algorithm 2) can be reduced while ensuring the termination conditions in line 18 of Algorithm 2.

V. SIMULATIONS AND DISCUSSIONS

In this section, simulation results are presented to show the IRR. The main parameters used in the simulations are summarized in Table I. Without additional explanations, the unit transmit power and energy costs are $c_s = 1$, $c_u = 10^{-4}$, $c_{jp} = 1$, and $c_j = 10^{-4}$. To guarantee a positive reward for the jammer under different scenarios, U^{up} is set as 6. A simple pre-conditioning is performed where the investigated problem is scaled such that the UAV altitude is 10 m.

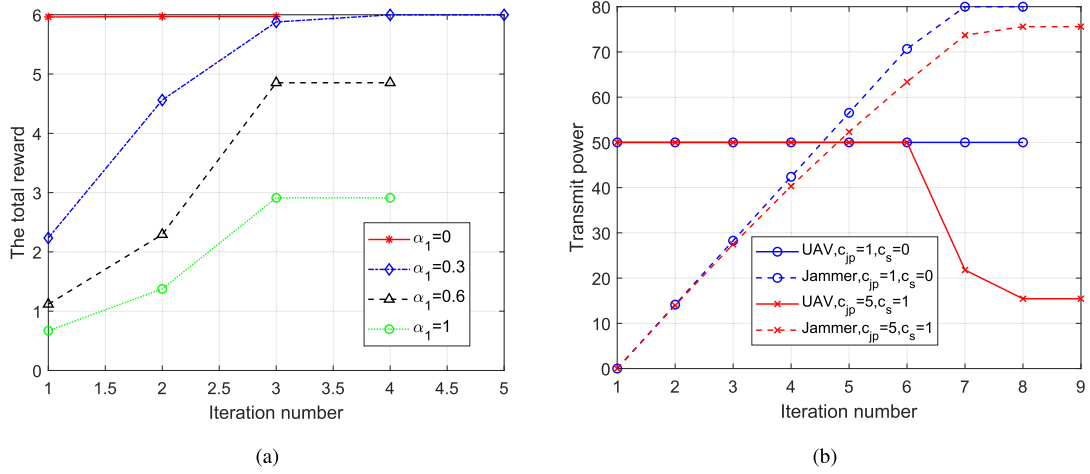


Fig. 3. Convergence behaviour of the proposed method. (a) The total reward with different rate-profile vectors $\mathbf{a}$, (b) The transmit power with different transmit cost (c_{jp}, c_s) .

TABLE I
MAIN PARAMETERS USED IN THE SIMULATIONS

Parameters	Values (Unit)
Maximum transmit power of the ground station P_s	50 mW
Maximum jamming power $\bar{P}_j$	80 mW
UAV altitude H	100 m
The pathloss exponent α	2.3
The channel power gain at the reference distance β_0	-60 dB
The Rician factor K_c	20
The noise power σ^2	-110 dBm
The number of time slots N	80
The time slot length δ_t	0.5 s
The location of the ground station $\mathbf{q}_s$	(0, 0) m
The start location of UAV $\mathbf{q}_u[0]$	(200, 200) m
The start location of jammer $\mathbf{q}_j[0]$	(300, 300) m
Maximum speed of UAV $V_u^{\max}$	30 m/s
Maximum speed of jammer $V_j^{\max}$	20 m/s

A. Reward Region Comparison With Power Control

For Algorithm 1 that is used to characterize the reward region with power control, we set the terminal condition as $(u_0^k - u_0^{k-1})/u_0^{k-1} \leq 10^{-3}$. Fig. 3(a) illustrates the convergence behaviour of Algorithm 1 under various parameters α . We can observe that the total reward increases with the number of iterations, and then converges to a stable value after several iterations. As expected, the optimal value is upper-bounded by U^{up} . The total reward u_0 decreases with increasing parameter α_1 , which reflects the weight value of UAV reward versus the total reward. Since the UAV reward U_u must be upper-bounded by a constant, the total reward is also limited by U_u and decreases with increasing α_1 considering $\alpha_1 u_0 \leq U_u$. Fig. 3(b) shows the transmit power obtained by Lemmas 1 and 2. It can be observed that the jamming power is maximized provided that $c_{jp} = 0$. This is intuitive because the jammer has no cost. However, as jamming and communicating costs increase, the jammer and UAV both reduce the power.

Fig. 4 shows the UAV and jammer rewards obtained by different methods under various transmit power costs, where SE-I and SE-II are the results obtained by the transmit power design schemes proposed in Section III-B, and SINR based game method refers to the anti-jamming transmission stackelberg game proposed in [35]. It can be first observed that all the

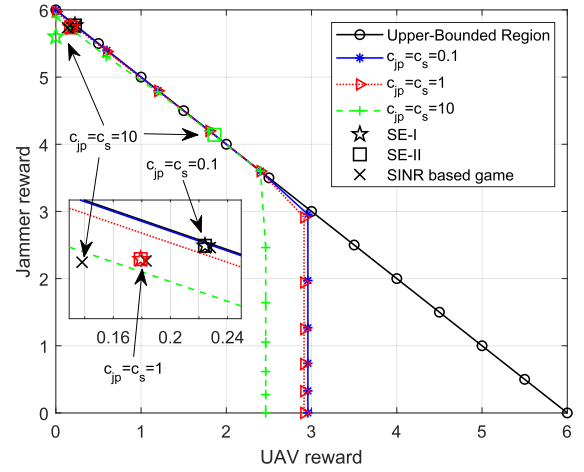


Fig. 4. IRR comparison under various settings, where SE-I and SE-II are the results obtained from game theoretic perspectives.

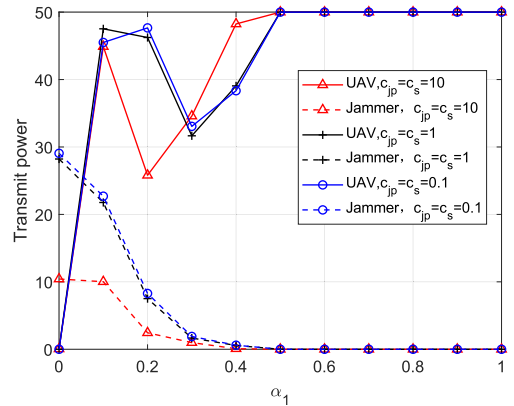


Fig. 5. Transmit power of the ground station and jamming power of the jammer versus the rate-profile vector.

reward pairs are limited by the line segment $u_1 + u_2 \leq U^{\text{up}}$. As the UAV reward improves, the jammer reward decreases. The maximum UAV reward is almost 3, and it will reduce with an increasing unit cost of transmit power (c_{jp}, c_s) . When

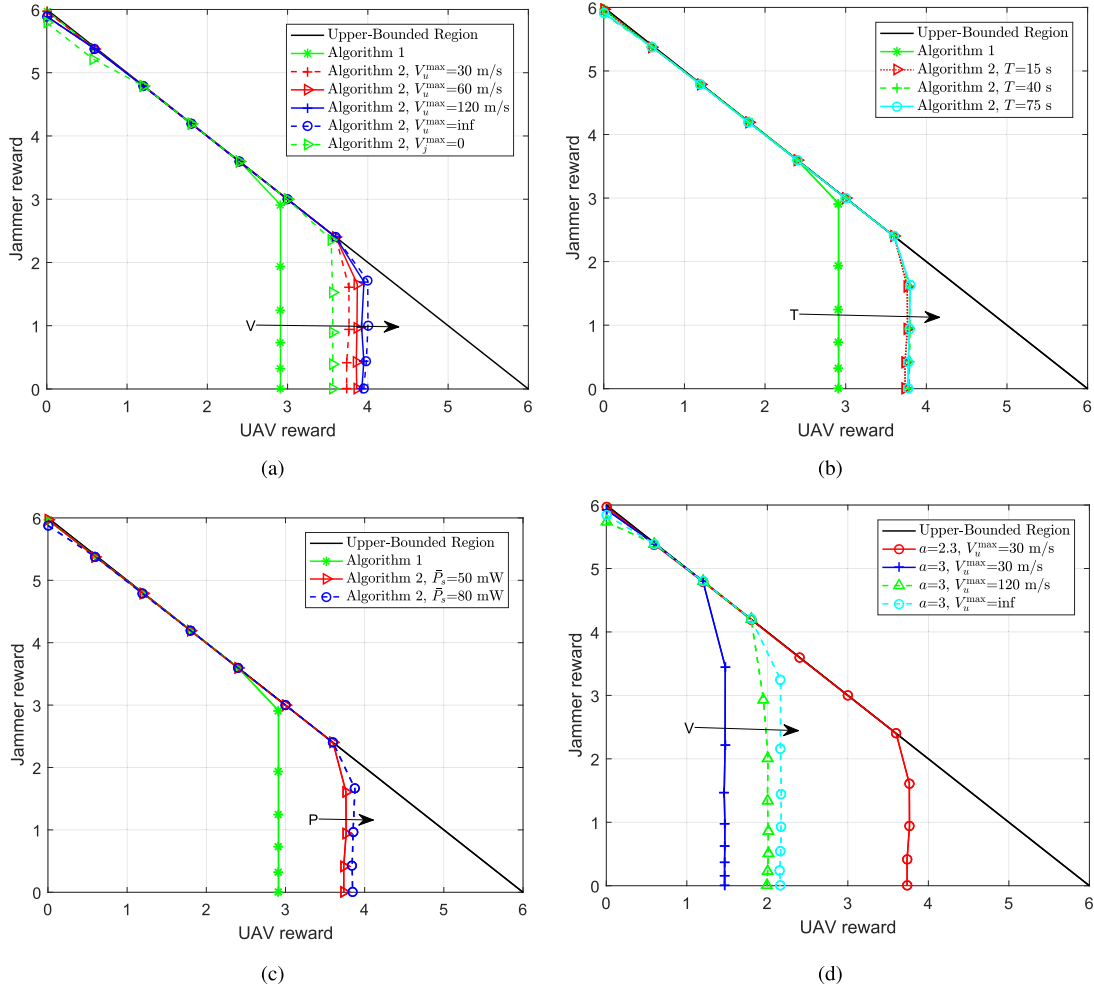


Fig. 6. Achievable reward regions with joint power and trajectory control under different maximum flight speeds V : (a) time lengths T , (b) maximum transmit powers P , (c) pathloss exponents a (d).

the UAV reward approaches to 3, the jammer reward drops from 3 to nearly 0. This is because the objective u_0 is limited by the maximum UAV reward. Thus, the jammer reward $\alpha_2 u_0$ will reduce with a decreasing value α_2 . It is noted that herein the jammer reward is calculated as $\alpha_2 u_0$ with rate profile method. In practical case, the jammer reward can be more than $\alpha_2 u_0$ by calculating $U_j = \frac{1}{N} \sum_n (U^{\text{up}} - U_u[n] - c_{jp} p_j[n])$, especially for $\alpha_1 \geq \alpha_2$. Since in the case of $p_j[n] = 0, \forall n$, there is $\max U_j = \max\{U^{\text{up}} - U_u, \alpha_2 u_0\}$. Considering $U_u \geq \alpha_1 u_0$, if $U^{\text{up}} \geq 2U_u$, then, there is $U^{\text{up}} - U_u \geq \alpha_2 u_0$ with $\alpha_1 \geq \alpha_2$.

The achievable reward pairs by Stackelberg game are also shown in Fig. 4. It can be observed that all the pairs are limited by the reward region achieved by Algorithm 1, and most pairs are concentrated on the upper-left corner in these settings, i.e., the jammer outperforms the UAV system. Moreover, the UAV (jammer) role in the game (i.e., leader or follower) cannot determine the obtained system performance. For example, in the case of $c_{jp} = c_s = 10$, the equilibriums for SE-I and SE-II are (0.04, 5.66) and (1.86, 4.14). Obviously, the case of UAV as the leader in the UAV reward is inferior to the case that the jammer acts as leader, and vice versa. Furthermore,

the achievable reward pairs obtained by three game-based approaches are almost the same with low costs. Note that the transmit power design schemes proposed in Section III-B are based on the throughput, while the transmission strategy proposed in [35] is based on the SINR at the legitimate receiver. With the increasing transmit power costs (i.e., $c_{jp} = c_s = 10$), the gap between three game-based approaches is larger.

Fig. 5 shows the transmit power of the ground station and jamming power of the jammer. As expected, the jamming power shows a downward trend as the ground station increases its transmit power. From a collaboration perspective, there is a cooperative trend between the UAV and jammer. In particular, the ground station communicates with the UAV with a lower transmit power if jamming power is high, in order to achieve better reward. On the other hand, the jammer reduces jamming power when the ground station increases its transmit power. From a confrontation perspective, when one increases its power, the other needs to pay a higher price to reduce rival's reward. Under this circumstance, lower transmit or jamming power is expected to obtain an acceptable reward.

B. Reward Region Comparison With Power and Mobility Control

To start the proposed Algorithm 2, we consider that UAV and jammer fly along with X-axis from the start points. Without loss of optimality, the power optimization (i.e., lines 5-9 in Algorithm 2) and path optimization (i.e., lines 12-16 in Algorithm 2) are iteratively implemented and only once at each iteration.

Fig. 6 presents the reward region that can be achieved by joint transmit power and path optimization. The reward region obtained by Algorithm 1 is also presented for comparison, where only transmit power is optimized in Algorithm 1. Fig. 6(a)-Fig. 6(d) compare the reward regions in terms of maximum flight speeds V , time lengths T , maximum transmit power P , pathloss exponents α , respectively. Unless otherwise specified, the maximum speeds of UAV and jammer are 30 m/s and 20 m/s. The maximum transmit power of ground station and jamming power of the jammer are 50 mW and 80 mW. The pathloss exponent is 2.3. It can be firstly observed from Fig. 6(a)-Fig. 6(d) that joint transmit power and path optimization can enlarge the reward region, especially for the UAV reward, which means that joint transmit power and path optimization achieves better anti-jamming performance. As expected, increasing the UAV mobility, flight time length, and transmit power of ground station can improve the UAV reward. The reason can be summarized as that all these factors provide more resource for the UAV against jamming from different aspects. In particular, the improvements of UAV flight speed make the UAV fly away from the jammer more quickly, thereby alleviating the jamming power. Similarly, the ground station can promote data transmissions by increasing the transmit power against jamming.

Comparing the cases that $V_j^{\max} = 20$ m/s and $V_j^{\max} = 0$ m/s, we can find that the UAV can achieve higher reward in the case of $V_j^{\max} = 20$ m/s, which is unusual in the countermeasure scenario where the UAV and jammer conflict with each other. Generally speaking, enhancing the ability of jammer will degrade the UAV reward. However, the aim of this paper is to characterize the reward region composed of all possible situations. In this context, the jammer can be misguided to assist the UAV for achieving higher reward, e.g., with zero jamming power or moving away from the UAV. These observations inspire us to design better anti-jamming strategy. For example, if the jamming power and trajectory of the jammer in which the UAV achieves the maximum reward is known, we can design some strategies to induce the jammer to perform specific actions.

The reward regions under different pathloss exponents are presented in Fig. 6(d), which also shows the impacts of flight speed. It can be seen that the increase of the pathloss exponent greatly shrink the reward region. This is because the channel conditions deteriorate, and thus the UAV reward decreases.

Fig. 7 gives the trajectories of UAV and jammer under different reward requirements. In all cases, the UAV tries to get close to the ground station. On the other hand, the UAV attempts to avoid the jammer via adjusting its spatial locations. In the cases of $\alpha_1 = 0.1$ and $\alpha_1 = 0.5$, the jammer always

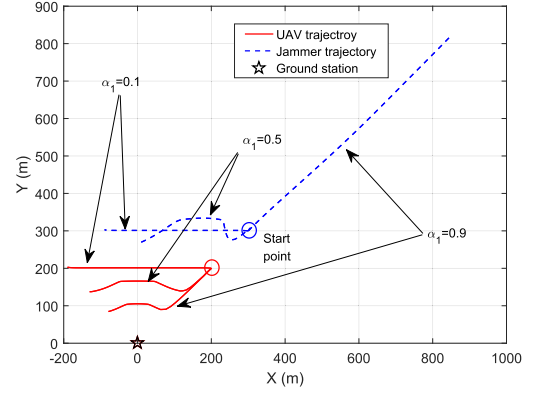


Fig. 7. Trajectory comparison of UAV and jammer under different reward requirements.

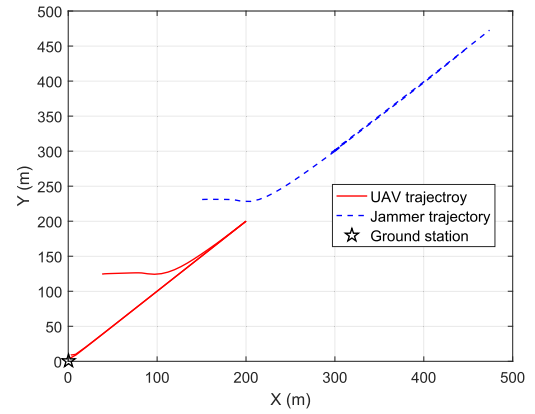


Fig. 8. Trajectory comparison of UAV and jammer under $a = 3$.

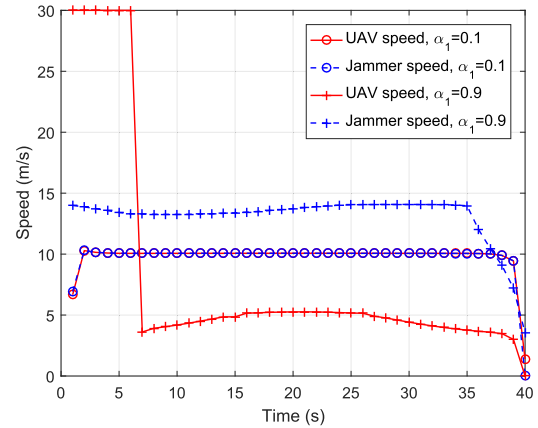


Fig. 9. Speed comparison of UAV and jammer under different reward requirements.

tracks the UAV in order to obtain higher rewards. In the case of $\alpha_1 = 0.9$, the jammer moves along with the opposite direction of the UAV flight to make the UAV achieve greater benefits. This phenomenon can be easily observed in Fig. 8 where the pathloss exponent $a = 3$. As predicted in Fig. 6, some results in Fig. 7 show that the jammer moves in the opposite direction of the UAV, which may benefit UAV communication. This is just one of all possible situations to characterize the IRR,

and the jammer may also stay close to the ground station. We further compare the trajectories of UAV and jammer versus pathloss exponent, which demonstrates that their trajectories have a great influence on communication performance under larger pathloss exponent case.

In order to further understand the movement behaviors of the UAV and jammer, Fig. 9 shows UAV and jammer speeds in the cases of $\alpha_1 = 0.1$ and $\alpha_1 = 0.9$. It can be seen from the figure that the UAV is expected to approach the ground station at the fastest speed (about 30 m/s), while reducing the flight speed (about 5 m/s) near the ground station. Accordingly, the jammer adjusts its speed to match the UAV.

VI. CONCLUSION

In this paper, we investigated the IRR characterization problem for the scenario that the UAV and jammer compete with each other to maximize their rewards via power optimization and trajectory planning. Two cases that the UAV and jammer are with limited mobility and high mobility were studied, respectively. Based on alternating optimization and SCA techniques, we developed the reward region characterization algorithms. Moreover, we presented two power control design schemes from a game-theoretic perspective, where the interactions between the UAV and jammer are formulated as a Stackelberg game. Simulation results were presented to show the reward regions under various settings. It was revealed that joint transmit power and path optimization can enlarge the reward region, while increasing the UAV mobility, flight time length, and transmit power improves the UAV reward. To realize better anti-jamming performance, suitable strategies can be carefully designed to induce the jammer to produce specific behaviors for improving the UAV reward;

APPENDIX A PROOF OF PROPOSITION 1

Under the case of $\{c_s, c_u, c_{jp}, c_j\} = \mathbf{0}$, the rewards of UAV and jammer are $U_u = \frac{1}{N} \sum_{n=1}^N R_{su}[n]$ and $U_j = \frac{1}{N} \sum_{n=1}^N (U^{\text{up}} - R_{su}[n])$. Then, the optimization is as follows;

$$\begin{aligned} & \max_{u_0, \{\mathbf{q}_u[n], \mathbf{q}_j[n], p_s[n], p_j[n]\}} u_0 \\ & \text{s.t. } \frac{1}{N} \sum_{n=1}^N R_{su}[n] \geq \alpha_1 u_0, \end{aligned} \quad (40a)$$

$$\frac{1}{N} \sum_{n=1}^N (U^{\text{up}} - R_{su}[n]) \geq \alpha_2 u_0, \quad (40b)$$

$$(8d), (8e). \quad (40c)$$

For any given feasible solution, there is

$$\begin{aligned} u_1 + u_2 &= \alpha_1 u_0 + \alpha_2 u_0 \leq \frac{1}{N} \sum_{n=1}^N R_{su}[n] \\ &+ \frac{1}{N} \sum_{n=1}^N (U^{\text{up}} - R_{su}[n]) = U^{\text{up}}. \end{aligned} \quad (41)$$

Therefore, the boundary is limited by the straight line $u_1 + u_2 = U^{\text{up}}$. Considering $u_1 \geq 0$ and $u_2 \geq 0$, the boundary of reward region is a triangle connecting three points $(0, 0)$, $(0, U^{\text{up}})$, and $(U^{\text{up}}, 0)$.

APPENDIX B PROOF OF LEMMA 1

From (16), we have

$$\begin{aligned} & \frac{\partial U_j(p_s, p_j)}{\partial p_j} \\ &= \frac{\beta_{ju} \log_2 e}{p_s \beta_{su} / (K_c + 1) + p_j \beta_{ju} + \sigma^2} - \frac{\beta_{ju} \log_2 e}{\gamma_2^k} - c_{jp}, \end{aligned} \quad (42a)$$

$$\begin{aligned} & \frac{\partial^2 U_j(p_s, p_j)}{\partial p_j^2} \\ &= \frac{-\beta_{ju} \log_2 e}{[p_s \beta_{su} / (K_c + 1) + p_j \beta_{ju} + \sigma^2]^2} \leq 0. \end{aligned} \quad (42b)$$

Thus, $U_j(p_s, p_j)$ is concave on p_j , and the optimal solution can be obtained by solving $\frac{\partial U_j(p_s, p_j)}{\partial p_j} = 0$. Let $\frac{\partial U_j(p_s, p_j)}{\partial p_j} = 0$, there is

$$\tilde{p}_j = \frac{1}{\beta_{ju} / \gamma_2^k + c_{jp} \ln 2} - \frac{p_s \beta_{su}}{(K_c + 1) \beta_{ju}} - \frac{\sigma^2}{\beta_{ju}}. \quad (43)$$

If $\tilde{p}_j \leq 0$, i.e., $p_s \geq \left(\frac{\beta_{ju}}{\beta_{ju} / \gamma_2^k + c_{jp} \ln 2} - \sigma^2 \right) \frac{K_c + 1}{\beta_{su}}$, then, U_j decreases with p_j , yielding $p_j = 0$. Similarly, if $\tilde{p}_j \geq \bar{P}_j$, i.e., $p_s \leq \left(\frac{\beta_{ju}}{\beta_{ju} / \gamma_2^k + c_{jp} \ln 2} - \beta_{ju} \bar{P}_j - \sigma^2 \right) \frac{(K_c + 1)}{\beta_{su}}$, then, U_j increases with p_j , yielding $p_j = \bar{P}_j$; otherwise, $p_j = \tilde{p}_j$. Lemma 1 is proved. ■

APPENDIX C PROOF OF LEMMA 2

$U_u(p_s)$ is a piecewise function, and it can be easily proved that $U_u(p_s)$ is concave with respect to p_s within each interval, $0 \leq p_s \leq p_s^1$, $p_s^1 \leq p_s \leq p_s^2$, $p_s^2 \leq p_s \leq \bar{P}$. Based on this observation, the following first-order derivative can be leveraged to obtain the optimal value of $U_u(p_s)$ within the feasible region,

$$\begin{aligned} & \frac{dU_u(p_s)}{dp_s} \\ &= \begin{cases} \frac{(K_c + 2) \beta_{su} \log_2 e}{(K_c + 2) p_s \beta_{su} + (K_c + 1) (\bar{P}_j \beta_{ju} + \sigma^2)} \\ \quad - \frac{\beta_{su} \log_2 e}{\gamma_1^k (K_c + 1)} - c_s, \Pi_1, \\ \frac{(K_c + 2) \beta_{su} \log_2 e}{(K_c + 2) p_s \beta_{su} + (K_c + 1) \sigma^2} \\ \quad - \frac{\beta_{su} \log_2 e}{\gamma_1^k (K_c + 1)} - c_s, \Pi_2, \\ \frac{\beta_{su} \log_2 e}{p_s \beta_{su} + \gamma_2^k \beta_{ju} / (\beta_{ju} + c_{jp} \gamma_2^k \ln 2)} - c_s, \text{otherwise.} \end{cases} \end{aligned} \quad (44)$$

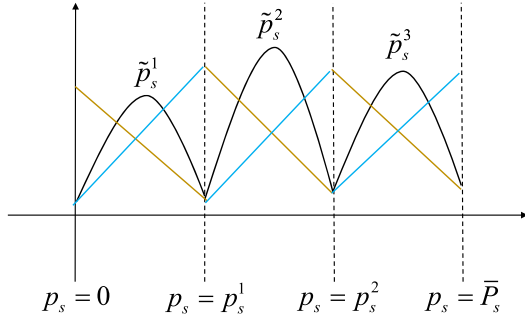


Fig. 10. Schematic diagram of $U_u(p_s)$ under various settings.

Without considering the power constraint, the optimal transmit power of UAV is derived at $\frac{dU_u(p_s)}{dp_s} = 0$. Then, we have

$$\tilde{p}_s = \begin{cases} \frac{(K_c + 1) \gamma_1^k}{\beta_{su} + (K_c + 1) \gamma_1^k c_s \ln 2} - \frac{(K_c + 1) (p_j \beta_{ju} + \sigma^2)}{(K_c + 2) \beta_{su}}, & \Pi_1, \\ \frac{(K_c + 1) \gamma_1^k}{\beta_{su} + (K_c + 1) \gamma_1^k c_s \ln 2} - \frac{(K_c + 1) \sigma^2}{(K_c + 2) \beta_{su}}, & \Pi_2, \\ \frac{\log_2 e}{c_s} - \frac{\gamma_2^k \beta_{ju}}{(\beta_{ju} + c_{jp} \gamma_2^k \ln 2) \beta_{su}}, & \text{otherwise.} \end{cases} \quad (45)$$

Within any interval, if $p_s \leq \tilde{p}_s$, then, there is $\frac{dU_u(p_s)}{dp_s} \geq 0$; otherwise, $\frac{dU_u(p_s)}{dp_s} \leq 0$. Therefore, $U_u(p_s)$ is first non-decreasing and then non-increasing within each interval, if conditions Π_1 and Π_2 are satisfied at $\tilde{p}_s$. If $\tilde{p}_s$ is not included in conditions Π_1 and Π_2 , then, $U_u(p_s)$ would be monotonic (non-decreasing or non-increasing) within an interval. Fig. 10 illustrates the schematic diagram of $U_u(p_s)$ under various settings. It can be observed that the optimal transmit power must be included by $p_s^* \in \{0, p_s^1, p_s^2, \tilde{p}_s^1, \tilde{p}_s^2, \tilde{p}_s^3, \bar{P}_s\}$ with $0 \leq \tilde{p}_s^1 \leq p_s^1 \leq \tilde{p}_s^3 \leq p_s^2 \leq \tilde{p}_s^2 \leq \bar{P}_s$. Thus, there is $p_s^* = \arg \max_{p_s \in \{0, p_s^1, p_s^2, \tilde{p}_s^1, \tilde{p}_s^2, \tilde{p}_s^3, \bar{P}_s\} \cap \bar{P}_s} U_u(p_s)$.

APPENDIX D PROOF OF LEMMA 3

When the jammer acts as leader, the jammer first selects the transmit power p_j and then the UAV makes corresponding decision p_s . With given transmit power p_j , it can be easily proved that the UAV's utility $U_u(p_s, p_j)$ is concave on p_s . Thus, the optimal transmit power can be obtained by setting $\frac{\partial U_u(p_s, p_j)}{\partial p_s} = 0$,

$$p_s = \begin{cases} \bar{P}_s, & \Pi_3 : p_j \leq p_j^1 = \frac{\gamma_1^k (K_c + 2) \beta_{su}}{\beta_{ju} \beta_{su} + c_s \gamma_1^k \beta_{ju} (K_c + 1) \ln 2} \\ & - \frac{\bar{P}_s (K_c + 2) \beta_{su}}{(K_c + 1) \beta_{ju}} - \frac{\sigma^2}{\beta_{ju}}, \\ 0, & \Pi_4 : p_j \geq p_j^2 = \frac{\gamma_1^k (K_c + 2) \beta_{su}}{\beta_{ju} \beta_{su} + c_s \gamma_1^k \beta_{ju} (K_c + 1) \ln 2} \\ & - \frac{\sigma^2}{\beta_{ju}}, \\ \frac{\gamma_1^k (K_c + 1)}{\beta_{su} + c_s \gamma_1^k (K_c + 1) \ln 2} - \frac{(K_c + 1) (p_j \beta_{ju} + \sigma^2)}{(K_c + 2) \beta_{su}}, & \text{otherwise.} \end{cases}$$

Substituting the UAV's transmit power in (16a) with optimal solution (46), there is

$$U_j(p_j) = \begin{cases} \log_2 \left(\frac{\bar{P}_s \beta_{su}}{(K_c + 1)} + p_j \beta_{ju} + \sigma^2 \right) - \left[\frac{\beta_{ju}}{\gamma_2^k \ln 2} + c_{jp} \right] p_j + b_2, & \Pi_3, \\ \log_2 (p_j \beta_{ju} + \sigma^2) - \left[\frac{\beta_{ju}}{\gamma_2^k \ln 2} + c_{jp} \right] p_j + b_1, & \Pi_4, \\ \log_2 \left(\frac{K_c + 1}{K_c + 2} p_j \beta_{ju} + b_3 \right) - \left[\frac{c_s (K_c + 1) \beta_{ju}}{(K_c + 2) \beta_{su}} + c_{jp} \right] p_j + b_4, & \text{otherwise,} \end{cases}$$

where $b_1 = U^{\text{up}} + \frac{\beta_{ju} p_j^k}{\gamma_2^k \ln 2} + \frac{(K_c + 2) \beta_{su}}{\gamma_2^k (K_c + 1) \ln 2} p_s^k - \log_2 \gamma_2^k$, $b_2 = b_1 - \frac{(K_c + 2) \beta_{su} \bar{P}_s}{\gamma_2^k (K_c + 1) \ln 2} + c_s \bar{P}_s$, $b_3 = \frac{\gamma_1^k \beta_{su}}{\beta_{su} + c_s \gamma_1^k (K_c + 1) \ln 2} + \frac{K_c + 1}{K_c + 2} \sigma^2$, and $b_4 = \frac{\gamma_1^k \gamma_2^k (K_c + 1) c_s - (K_c + 2) \gamma_1^k \beta_{su} \log_2 e}{\beta_{su} \gamma_2^k + c_s \gamma_1^k \gamma_2^k (K_c + 1) \ln 2} - \frac{c_s (K_c + 1) \sigma^2}{(K_c + 2) \beta_{su}} + \frac{\sigma^2}{\gamma_2^k \ln 2} + b_1$. We can further get the first-order derivative with respect to p_j as follows,

$$\frac{dU_j(p_j)}{dp_j} = \begin{cases} \frac{\beta_{ju} \log_2 e}{\bar{P}_s \beta_{su} / (K_c + 1) + p_j \beta_{ju} + \sigma^2} - \left(\frac{\beta_{ju}}{\gamma_2^k \ln 2} + c_{jp} \right), & \Pi_3, \\ \frac{\beta_{ju} \log_2 e}{p_j \beta_{ju} + \sigma^2} - \left(\frac{\beta_{ju}}{\gamma_2^k \ln 2} + c_{jp} \right), & \Pi_4, \\ \frac{(K_c + 1) \beta_{ju} \log_2 e}{(K_c + 1) p_j \beta_{ju} + (K_c + 2) b_3} - \left(\frac{c_s (K_c + 1) \beta_{ju}}{(K_c + 2) \beta_{su}} + c_{jp} \right), & \text{otherwise.} \end{cases} \quad (46)$$

Then, the jammer's optimal transmit power in the absence of power constraints is

$$p_j^* = \begin{cases} \frac{\gamma_2^k}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} - \frac{\bar{P}_s \beta_{su}}{\beta_{ju} (K_c + 1)} - \frac{\sigma^2}{\beta_{ju}}, \\ \frac{\gamma_2^k}{\beta_{ju} + c_{jp} \gamma_2^k \ln 2} - \frac{\sigma^2}{\beta_{ju}}, \\ \frac{(K_c + 2) \beta_{su} \log_2 e}{c_s (K_c + 1) \beta_{ju} + c_{jp} (K_c + 2) \beta_{su}} - \frac{(K_c + 2) b_3}{(K_c + 1) \beta_{ju}}. \end{cases} \quad (47)$$

Therefore, in the presence of power constraints $\tilde{P}_j = \{0 \leq \tilde{p}_j^1 \leq p_j^1 \leq \tilde{p}_j^3 \leq p_j^2 \leq \tilde{p}_j^2 \leq \bar{P}_j\}$, the optimal transmit power is $p_j^* = \arg \max_{p_j \in \{0, p_j^1, p_j^2, \tilde{p}_j^1, \tilde{p}_j^2, \tilde{p}_j^3, \bar{P}_j\} \cap \bar{P}_j} U_j(p_j)$.

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